

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

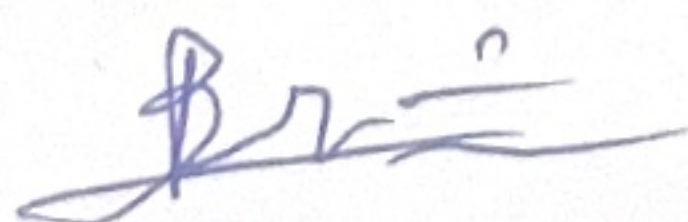
Annexure A

Analytical Problem Solving

Code of Conduct:

(192573016)
I M. PRASANNA VENKATESH (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Jan 12 - 3
EP

Solve the 6-Queens Problem

The 6-Queens problem is a well-known problem in Artificial Intelligence and search algorithms. The objective is to place six queens on a 6×6 chessboard so that no queen attacks another queen. A queen in chess can move horizontally, vertically and diagonally. Therefore, no two queens should be in the same row, same column, or same diagonal.

The 6-Queens problem is solved using state-space search and backtracking, where invalid arrangements are discarded and only safe positions are explored. It demonstrates how AI can solve constraint satisfaction problems efficiently.

Problem components

1. State space

The state space consists of all possible arrangements of queens on a 6×6 chessboard. Each state represents the positions of queens placed on the board. Since there are many possible arrangements, the state space is very large.

2. Initial state

The Initial state is an empty 6×6 chessboard with no queens placed.

3. Successor Function

The successor function generates the next possible state by placing one queen in the next row. Only positions that

do not conflict with already placed queens are considered.

4. Goal Test

The goal is reached when all six queens are placed successfully without attacking one another. If all constraints are satisfied the algorithm stops and returns the solution.

Incremental Formulation

In incremental formulation, queens are placed one at a time. After placing each queen, the algorithm immediately checks whether it attacks any previously placed queen.

If a conflict occurs:

- The current position is rejected
- The algorithm backtracks
- Another position is tried

Complete - state Formulation

In complete - state formulation, all six queens are placed on the board first. After the complete arrangement is formed, the algorithm checks whether the queens attack each other.

This approach is inefficient because it evaluates many invalid arrangements before finding a valid solution.

Advantages

- Reduces unnecessary search
- Saves execution time
- Requires less memory
- Suitable for constraint satisfaction problems.

Chess - Playing Agent Environment

A chess-playing agent is an Artificial Intelligence system designed to play chess against a human or another computer. The agent observes the board, analyses possible moves, predicts future outcomes, and selects the best move. According to AI, the chess environment is classified using six environment properties.

1. Fully Observable
Chess is a fully observable environment because every piece on the board is visible to both players. There is no hidden information. The agent knows the exact position of every piece before making a decision.

Justification:
* Complete information is available
* No uncertainty about the board
* Every legal move can be analysed

2. Deterministic
Chess is deterministic because every legal move has a predictable result. There is no randomness involved.

Justification:
* A move always produces the same outcome.
* Dice or chance are not used
* The next state depends only on the chosen move.

3. Sequential

Chess is a sequential environment because every move influences future moves. Decisions cannot be made independently.

Justification:

- * One move changes the game
- * Future strategies depend on previous moves
- * Long-term planning is essential.

4. Static

Chess is considered static because the board does not change while a player is thinking.

Justification:

- * Pieces remain in their positions
- * No automatic movement occurs
- * Players can analyse without environmental changes

5. Discrete

Chess is a discrete environment since it has a finite number of board positions and legal moves.

Justification:

- * The board contains 64 squares
- * Moves occur one at a time
- * States are clearly defined

6. Multi-Agent

Chess is a multi-agent environment because two intelligent players compete with each other.

Justification:

- * Each player has different goals
- * One player's move affects the opponent
- * Both players continuously adapt strategies

Importance of AI in chess

AI uses algorithms such as:

- Minimax Algorithm
- Alpha - Beta Pruning
- Heuristic Evaluation
- Game Tree search

3) Exploration Robot in an Unknown Cave
An exploration robot is sent into an unknown cave without any map. The robot has limited sensors, meaning it cannot observe the entire cave at once. Its objective is to safely explore the cave, build a map, avoid dangerous areas and finally locate an exit.

online (contingency) search

Online search means the robot plans while acting. Instead of having complete information before starting, it collects information as it moves.

The robot:

- * observes nearby areas
- * chooses the safest next move
- * updates its knowledge continuously
- * changes its plan whenever new information is discovered.

Building the Map

As the robot explores:

- * It stores visited locations
- * Marks unexplored passages
- * Identifies obstacles
- * Detects dead ends
- * Updates the map after every movement

Searching for the Exit

The robot searches systematically by:

- * Exploring one path at a time
- * Returning from dead ends
- * Choosing unexplored routes
- * Updating its search strategy.

Risk of Irreversible Actions

Irreversible actions are dangerous because they cannot be undone.

Examples include:

- * Entering a narrow passage with no return path
- * Falling into pits
- * Crossing unstable bridges
- * Running out of batteries.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

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